

dimmed down to zero.

A function that allows dimming to 10 or 20 per cent is, however, not satisfactory, as conventional incandescent lamps can be dimmed to much lower levels. In addition, the colour temperature of a dimmed incandescent lamp is shifted to much warmer levels, while LEDs show no such shift. The 10 per cent brightness of an LED luminaire powered with a residual current is therefore perceived by the eye as much higher, equivalent to about 35 per cent brightness of an incandescent lamp. Dimming to levels below 5 per cent is thus even more crucial for LED lighting systems.

In the recent past, LEDs have also been developed further at a rapid pace. Initially, several individual 2W or 3W LEDs with separate housings were combined on a PCB. Today, the trend is clearly towards multi-chip solutions. For this purpose, a number of small LED dies are mounted on ceramic chips. The ceramic substrate improves heat management across the entire LED array (Fig. 3).

In addition, such LED arrays require much less space, and the entire luminous surface is covered by a phosphorus coating so that the multi-chip LEDs appear as one single light source. This facilitates the design of reflective and optical devices.

This can be well illustrated by the

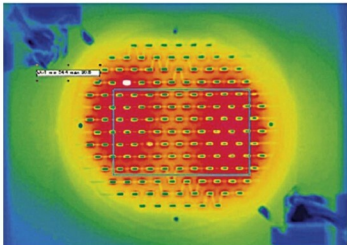


Fig. 3: Thermal image shows relatively homogeneous heat distribution in MegaZenigata (Image courtesy: Sharp)



Fig. 4: The new MegaZenigata from Sharp (left) contains 168 individual LEDs combined in an array (in series and parallel) powered by approx. 38V DC and a constant current of 700 mA (Image courtesy: Sharp)

25W MegaZenigata from Sharp: a total of 168 LEDs arranged over an area of just below 2 cm² (Fig. 4) are wired to form an array. This array can then be mounted on the MegaZenigata, using a specially devised LED driver.

The LED driver provides a constant-current output of 700 mA up to 42V, so that the MegaZenigata produces 2600 lumens at 4000°K, corresponding to the luminous flux of a 150W halogen spotlight. Of course,

the quality of the light plays a major role. While daylight reaches a colour rendering index of 100, the MegaZenigata achieves a respectable 83. The MegaZenigata is thus not only efficient but also offers a light quality and a colour temperature close to that of natural light.

Summary

In the future, manufacturers of LED drivers will cooperate even more closely with LED chip producers in order to take full advantage of the possibilities of new lighting technology. While energy efficiency and long service lives remain the main concerns, quality of the light is also a major issue, since it determines how we perceive the light.

The trend towards LED lighting systems will bring billions of new drivers into the global market over the next few years, which all need to be connected to

main networks. Because they will all produce harmonics and reactive currents, drivers need to have not only a high efficiency rating but also a good power factor. Values around 95 per cent should therefore be considered long-term guide values, even if they are not yet required by the relevant standardisation organisations. ●

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